# AQA GCSE Chemistry

# Topic 1: Atomic Structure and the Periodic Table

## A simple model of the atom, symbols, relative atomic mass, electronic charge and isotopes

### Notes

(Content in bold is for Higher Tier only)

## *Atoms, elements and compounds*

- Atoms make up all substances and are the smallest part of an element that can exist
- Chemical symbols represent an atom of an element e.g. Na represents an atom of sodium
- Compounds are formed from elements by chemical reactions. Chemical reactions always involve the formation of one or more new substances, and often involve an energy change
- Compounds contain two or more elements chemically combined in fixed proportions and can be represented by formulae using the symbols of the atoms from which they were formed e.g. HCl is a compound containing 1 atom of hydrogen and 1 of chlorine per molecule
- Compounds can only be separated into elements by chemical reactions

## *Mixtures*

- A mixture consists of two or more elements or compounds not chemically combined together. The chemical properties of each substance in the mixture are unchanged. (this is different to a compound)
- Can be separated by: filtration, crystallisation, simple distillation, fractional distillation and chromatography. These are physical processes, so do not involve chemical reactions and no new substances are made.


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# *Separating mixtures summary table:*

| Method | Use | Apparatus | Example/s |
|---|---|---|---|
| Filtration | Separating an insoluble solid from a liquid.<br><br>The liquid passes through the filter paper while the solid is left behind as residue. | Funnel, filter paper, beaker/s | Separating sand (residue) from water (filtrate)<br><br>Separating excess reactant (residue) from liquid product (filtrate) |
| Simple Distillation | Separating a solvent (liquid in which the solute dissolves) from a solution.<br><br>Used to separate a solvent from a solution/ liquids with very different boiling points. The lower boiling point solvent evaporates first and condenses separately. | Bunsen burner/ heating equipment, condenser, collection beaker/ flask | Separating pure water from sea water |
| Fractional Distillation | Separating two or more miscible (mixable) liquids that have different boiling points.<br><br>The mixture is heated so the liquids vapourise and rise up a column that is hot at the bottom and cold at the top.<br><br>The liquid with the lowest boiling point evaporates to the top of the column first, condenses, and is collected. Different liquids can be collected separately. | Fractionating column, thermometer, condenser, collection beaker, heating equipment | Separating ethanol from a mixture of ethanol and water |
| Paper Chromatography | Separates mixtures of soluble substances.<br><br>Relies on the substances having different affinities to the paper and the solvent. | Beaker, chromatography paper | Separating coloured components in ink |
| Crystallisation | Separates a dissolved solid from a solution.<br><br>Solution is warmed so that the solvent evaporates and slow cooling lets solid crystals of the solute form. | Evaporating basin, water bath, Bunsen burner/ heating equipment | Forming copper sulfate crystals from dilute sulfuric acid and copper oxide mixture. |

## *The development of the model of the atom*

- First: atoms were thought to be tiny spheres that could not be divided
- Discovery of electron-> plum pudding model (atom is a ball of positive charge with negative electrons embedded in it)
- Alpha particle scattering experiment-> conclusion that the mass of an atom was concentrated at the centre (nucleus) and that the nucleus was charged
  Scattering experiment:
  - A beam of alpha particles was aimed at very thin gold foil and their passage through was detected
  - Some of the alpha particles emerged from the foil at different angles, and some even came straight back
  - the positively charged alpha particles were being repelled and deflected by a small concentration of positive charge in the atom (nucleus)
- Neil Bohr: suggested electrons orbit the nucleus at specific distances (supported by experimental data)
- Later experiments: positive charge of any nucleus could be subdivided into a whole number of smaller particles, each particle having the same amount of positive charge (protons)
- James Chadwick's work: provided the evidence to show the existence of neutrons within the nucleus (had been an accepted scientific idea for about 20 years already)


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## *Relative electrical charges of subatomic particles*

- atomic number: the number of protons in an atom of an element.
- All atoms of a particular element have the same number of protons.
- Atoms of different elements have different numbers of protons.

| particle | Relative charge |
|---|---|
| proton | +1 |
| neutron | 0 |
| electron | -1 |

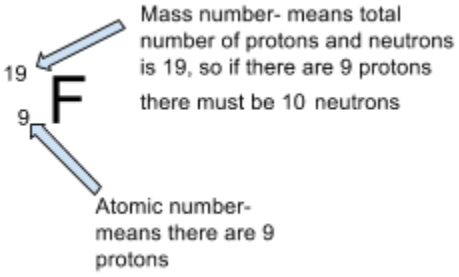
- An atom has an overall charge of 0, so number of protons = number of electrons

## *Size and mass of atoms*

- Atoms are very small ( radius of about 0.1 nm ) and the radius of a nucleus is less than 1/10,000 of that of the atom, though it holds almost all of the mass

| particle | Relative mass |
|---|---|
| proton | 1 |
| neutron | 1 |
| electron | Very small |

- Mass number: the sum of the protons and neutrons in an atom
- Isotopes: atoms of the same element with different numbers of neutrons
- This is how atoms are shown in the periodic table:




## *Relative atomic mass*

- Relative atomic mass: an average value that takes account of the abundance of the isotopes of the element

- Example question: carbon has 2 isotopes: carbon-14 with abundance 20% and carbon-12 with abundance 80%. Calculate the relative atomic mass of carbon.
  To calculate it: ((isotope 1 mass x abundance) + (isotope 2 mass x abundance)) ÷ 100
  For this question: ((14 x 20) + (12 x 80)) ÷ 100 = 1240 ÷ 100 = 12.4

## *Electronic structure*

- electrons occupy the lowest available energy levels (the shells closest to the central nucleus)
- Electronic structure of an atom tells you how many electrons are in each shell
  e.g. for sodium: 2 electrons in shell 1 (closest to nucleus), 8 in shell 2, 1 in shell 3
  Electronic structure= 2,8,1 or

